



SNAP AND SLIP

If rock strata are stretched or squeezed by the movements that cause earthquakes, they may snap. The result is a fault. The strata slip along the line of the fault, relative to each other, so layers that were joined together become offset. However, the nature and thickness of each layer usually enables them to be identified, even if they have slipped over a very long distance.



FOSSIL MARKERS

The sequence of rock strata can be mixed up by folds and faults, so it is not always obvious how old the rocks are. But in the 18th century, geologist William Smith saw that if rocks in different places contained the same fossils, the rocks were the same age. This allowed geologists to create geological maps, such as this one of the Grand Canyon, where each color shows a different rock age.



RADIOMETRIC DATING

Many elements in rocks are unstable. Their atoms decay over time, emitting radiation. Scientists know how long it takes for an element to decay. Uranium, for example, decays into lead at a steady rate. This radioactive dating technique has enabled geologists to put dates to all the eras and periods in the geological timescale.

